

Quorum at every scale: diagonal silence and surviving coupling

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September 1, 2026 — lemma campaign, item 3

Abstract

The certified quorum theorem is a statement at three fixed windows (340 witnesses). What remains is the statement at every scale: every proper sub-Euler product of the windowed Weil form is indefinite, for every μ . The reduction already measured at $\mu = 11$ cuts this to two harmonic-analysis lemmas on the radical — (B1) diagonal silence before recruitment, (B2) a surviving off-diagonal coupling — plus the observation that the radical dimension grows like the ladder. Neither lemma mentions RH. We state both, record what is certified, and isolate the one estimate that still scales.

1 What is already a theorem

On the 47-dimensional even window V at $\mu = 11$, and on the analogous windows at $\mu = 16$ and $\mu = 22$ (and for χ_3 at $\mu = 11$), every proper partial form Q_S admits an explicit negative witness, ball-certified [1]. That is 340 cases, no exception. It is a theorem about explicitly defined matrices. It is not a theorem about every μ .

At $\mu = 11$ a further reduction is measured exact [2] §15: the violator of $Q_{\setminus p}$ lives 99.4% in the quasi-radical of the complete Q , and

$$\lambda_{\min}(Q_{\setminus p}) = \lambda_{\min}(M_p) + O(10^{-4}),$$

where M_p is the tower of p restricted to the radical. Each M_p has a certified negative 2×2 minor (margins 0.10–0.19 against quadrature $< 10^{-8}$). The pattern of those minors is the content: a *silent* rung (diagonal ≈ 0 : p not yet recruited) paired with a *speaking* rung, off-diagonal of size ~ 0.4 . A matrix $\begin{pmatrix} \varepsilon & b \\ b & a \end{pmatrix}$ is indefinite as soon as $b^2 > \varepsilon a$, which is automatic when $\varepsilon \approx 0$.

2 The two lemmas

Let $v_0, v_1, \dots$ be the quasi-null eigenfunctions of the complete form Q at a window $\mu = e^L$, and write $k(p)$ for the first rung that recruits p (so $k(p) = n - 1$ when p is the n -th χ -supported prime, empirically). Write $(f \star g)(x) = \Theta_{f,g}(x)$ for the symmetrized correlation.

Lemma 1 (B1, diagonal silence). *For every interior prime p and every rung $j < k(p)$,*

$$(v_j \star v_j)(\log p) = o_p(1).$$

Equivalently the diagonal entry of M_p on a pre-recruitment rung vanishes.

Lemma 2 (B2, surviving coupling). *For every interior prime p there is at least one speaking rung k such that*

$$(v_j \star v_k)(\log p) \neq 0 \quad \text{for some silent } j < k(p).$$

The coupling is $O(1)$, not $o(1)$: it does not shrink with μ .

Conjecture 1 (Quorum at every scale). *B1 + B2 + the radical reduction $\Rightarrow Q_S$ is indefinite for every proper $S \subsetneq P(\mu)$ and every μ large enough that V is depth-adequate. No appeal to RH.*

Lemma A of [1] (every single tower is indefinite on V) is elementary and already proved; it does not locate the negative directions. B1 and B2 do.

3 What scales, what does not

The 2×2 certificate is dimension-free: once a silent/speaking pair exists, indefiniteness of M_p does not care how large the radical is. The radical dimension itself grows — $\ell_0 = s\mu$ produces $\sim s\mu/\Delta_\infty$ rungs — so B1 and B2 have to be checked against a growing flag, not a fixed 8×8 block.

The estimate that still scales is the reduction $\lambda_{\min}(Q_{\setminus p}) = \lambda_{\min}(M_p) + O(\varepsilon(\mu))$. At $\mu = 11$, $\varepsilon \sim 10^{-4}$ against $\lambda_{\min}(M_p) \sim -0.5$. As μ grows the complete form sits deeper ($\lambda_{\min}(Q) \sim e^{-s\mu}$) and the bulk gap above the ladder stays $O(1)$. A perturbation argument on that gap gives $\varepsilon(\mu) \rightarrow 0$, which is the friendly direction. Writing the $\varepsilon(\mu)$ bound in ball arithmetic at one more window ($\mu = 16$, radical already computed by the quorum engine) would promote the reduction from “measured exact at 11” to “certified in a range”.

B1 is the microscopic content of the recruitment law. A proof of B1 is a proof that $v_j \star v_j$ is supported, at leading order, on the logs of the primes already recruited by rung j . That is a statement about the correlation of windowed eigenfunctions at a discrete set of lags — harmonic analysis on the window, plus the seating of primes, not RH.

B2 is the assertion that silence is *diagonal only*. The off-diagonal $(v_j \star v_k)(\log p)$ is an oscillatory overlap with no symmetry that forces a zero. One non-vanishing pair per p suffices.

4 A finite check that would change the status

At $\mu = 16$ the quorum engine already built every Q_S . Restricting the four simple-deletion violators to the quasi-radical of the complete form and certifying one 2×2 minor per prime would repeat the $\mu = 11$ lemma-B campaign at a second scale. If the silent / speaking pattern recurs, B1+B2 stop being a $\mu = 11$ anecdote.

5 Status

The quorum at three windows is a theorem. B1 and B2 are the lemmas for every scale. The reduction $\varepsilon(\mu)$ is the one analytic estimate left. None of the three mentions RH.

References

- [1] D. Joubert, *Proper sub-Euler products violate Weil positivity on a fixed window: certified witnesses*, same repository (2026).
- [2] D. Joubert, *Le milieu des nombres premiers*, lab notebook, same repository (2026), §14.7–15.2.

Addendum — $\mu = 16$ witnesses, 1 September

The shipped file `witnesses_zeta_mu16.json` already certifies every proper Q_S negative at this window (theorem). Overlap of each simple-deletion violator with the complete-form ground state alone is 0.04–0.40 — not 0.99 as at $\mu = 11$ against the six-dimensional radical. That is expected: the radical has grown. The 2×2 minor campaign at $\mu = 16$ still needs the complete eigenspace, which is not an artifact. B1+B2 remain lemmas; the cheap lock that would promote them is “assemble Q at $\mu = 16$, project T_p on the first eight eigenvectors, certify one minor per interior prime.”

Addendum — B1+B2 at a second scale, 1 September

`code/lemma_B_chi.py` on χ_{-8} at $\mu = 16$ ($N = 29$, depth-adequate: $\ell = (16.27, 4.56)$). Interior primes $\{3, 5, 7, 11, 13\}$. Every tower, restricted to the first eight eigenvectors of the complete form, has a negative 2×2 minor:

p	silent j	pair	$\det_2$	B1+B2
3	6	(6,7)	-0.191	yes
5	3	(0,1)	-0.217	yes
7	0	(0,1)	-0.081	yes
11	0	(2,3)	-0.045	yes
13	0	(1,5)	-0.032	yes

Five for five. The same pattern as ζ at $\mu = 11$, on a different L -function, at a different window. B1+B2 are no longer a $\mu = 11$ anecdote. The $\varepsilon(\mu)$ reduction for ζ at $\mu = 16$ still wants a depth-adequate basis ($N \sim 71$), which this lock does not replace.